

DockFlow

Results, Technical Work and Impact

Clothing Warehouse & Delivery Management System



Sec No-13 | Team No-14

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How this review covers every criterion

1

Accuracy of Results

Slides 3-4

How results are checked
and what we measured

2

Technical Competence

Slides 5-6

Tools, design skills and
problems solved

3

Innovative & Sustainable

Slide 7

What is new and what
keeps it lasting

4

Utility to Society

Slide 8

Who benefits and SDG
alignment

5

Presentation Skills

Slides 1-9

Clear flow, visuals and
summary

REVIEW 4 TOTAL: 5 criteria x 10 marks = 50

How we check that DockFlow gives correct results

What must be accurate	How we check it	Measure
Stock counts	Add and remove known quantities, then compare with the counts shown	Mismatches found
Orders against stock	Create orders and check that stock reduces by exactly the ordered amount	Mismatches found
Delivery status	Update a shipment and check the status shown for that order	Correct status shown
Shipment events	Send events through Kafka; check MongoDB receives each one, repeats included	Events saved correctly
Search by size, color, location	Look up known items for men, women and kids	Correct item returned

Test data is known in advance, so every result can be marked right or wrong.

Results against our targets

Measure	Target	Test runs	Measured result
Stock counts matching real stock	100%	Add runs	Add result
Orders reducing stock correctly	100%	Add runs	Add result
Delivery status shown correctly	100%	Add runs	Add result
Shipment events saved in MongoDB	Every event	Add runs	Add result
Items found by size, color, location	100%	Add runs	Add result
Response time: stock and order actions	Under 1 second	Add runs	Add result

Results come from running these checks on the working system. Limits of testing are stated openly.

Technical skills behind DockFlow

1

MySQL: stock and orders

Relational tables for clothes by category, size, color and location, linked to orders so counts stay consistent.

2

MongoDB: delivery updates

Flexible documents for shipment updates, so new delivery details need no table change.

3

Kafka: shipment events

Events pass from the order side to the delivery side, keeping the two services separate.

4

Staff website

One browser interface to add stock, create orders and update deliveries.

Skills shown: database design, NoSQL modelling, event-driven design, and web development.

Technical problems and how we solved them

Challenge	Why it happens	How DockFlow handles it
Data split across two stores	Orders are in MySQL, delivery updates in MongoDB	Records linked by an order reference
	Kafka delivers at-least-once, so an event can arrive twice	Delivery updates are safe to repeat
	Each tool needs its own setup and care	One narrow job per tool
	Orders and stock change together	Orders created against tracked stock in MySQL

Honest limits: real courier GPS, payments, CI/CD and full C4 diagrams are future work.

What is innovative and what makes it sustainable

INNOVATIVE

Right database for each job

MySQL for stock and orders, MongoDB for delivery updates

Event-driven shipments

Kafka passes shipment events between services

One place for everything

Stock, orders and deliveries in one website

Flexible delivery records

New delivery details added without changing tables

SUSTAINABLE

Open-source stack

No paid licences, so it is affordable for small businesses

Less waste

Accurate stock reduces wrong, lost or surplus items

Adapts locally

New categories, fields or services change only one part

Less paperwork

Digital records replace manual re-entry

Who benefits, and how DockFlow fits the SDGs

Who	Need today	What DockFlow gives
Warehouse staff	Separate records for stock, orders and deliveries	One place for all three
Clothing businesses	Wrong stock counts and delayed orders	Accurate stock and smoother orders
Customers: men, women, kids	Wrong items and unclear delivery status	Right clothes with clear status
Delivery partners	Little shipment information shared	Clear shipment updates between services

SDG 9

Industry, Innovation & Infrastructure

Modern database tools for small logistics

SDG 12

Responsible Consumption & Production

Fewer wrong, lost or surplus items

SDG 8

Decent Work & Economic Growth

Less re-entry, more time for real work

Summary: DockFlow Review 4

Accurate

Five checks with known test data; results against targets.

Technically sound

MySQL, MongoDB and Kafka, each with one narrow job.

Innovative & sustainable

Right store per job; open-source; adapts locally.

Useful to society

Helps four groups; supports SDGs 9, 12 and 8.

Thank you

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We are happy to take your questions.